

MICHIGAN STATE UNIVERSITY

12 September 2026

Wellcome Leap
Resistance Networks Program

Re: Support for "Universal properties of plasmid evolution and robust risk forecasting for preemptive pARG intervention"

Dear Members of the Review Committee:

I write in support of the proposal submitted by Dr. Philip J. Gerrish to the Wellcome Leap Resistance Networks program.

I served as Phil's doctoral advisor at Michigan State University. During that period, and in some subsequent work, we've collaborated on studies of adaptation in microbial populations. Our joint publications include papers in *Nature* and *Science* on the evolution of high mutation rates and the diminishing returns from increased mutation supply, as well as theoretical work that on the phenomenon of clonal interference in asexual populations. These contributions combined mathematical analyses with results from microbial evolution experiments.

The present proposal builds directly on this background. Its treatment of antibiotic-resistance plasmids as evolving and transmissible genetic elements brings together several subjects central to Phil's research: mutation, selection, linkage, competition among lineages, population bottlenecks, and the interaction between vertical and horizontal transmission. The proposal's emphasis on quantitative predictions and prospective validation is appropriate for determining whether general evolutionary results can be used to improve the assessment and management of risks associated with antibiotic resistance.

Phil has demonstrated originality, independence, and persistence throughout his scientific career. He is particularly skilled at identifying general mechanisms within complex evolutionary processes and formulating them in ways that can be tested against data. In my judgment, he is very well qualified to lead this project and to coordinate its theoretical and empirical components.

In summary, I enthusiastically support this creative and ambitious proposal.

Sincerely,



Richard E. Lenski
University Distinguished Professor
and
Member, National Academy of Sciences, USA



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